PanelAppRex aggregates disease gene panels and facilitates sophisticated search

Quant Group[1], Simon Boutry[2], Ali Saadat[2], Sinisa Savic[3], Luregn J. Schlapbach[4], Jacques Fellay[2], and Dylan Lawless * [4]

[1]The quantitative omic epidemiology group.
[2]Global Health Institute, School of Life Sciences, École Polytechnique Fédérale de Lausanne, Switzerland.
[3]Leeds Institute of Rheumatic and Musculoskeletal Medicine, University of Leeds, Leeds, UK.
[4]Department of Intensive Care and Neonatology, University Children's Hospital Zurich, University of Zurich, Zurich, Switzerland.

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*Addresses for correspondence: Dylan.Lawless@kispi.uzh.ch

Abstract

Motivation:
Gene panel data are essential for variant interpretation and genomic diagnostics, but existing resources are fragmented, inconsistently annotated, and not easily accessible for programmatic use. We developed PanelAppRex, a harmonised dataset and interactive search tool that integrates over 58,000 curated gene-disease panel associations. It supports natural language-style queries by gene, phenotype, disease group, and mode of inheritance (MOI), with results returned in machine-readable export formats.

Results:
In benchmarked case studies of immune disease diagnostics, the system achieved 100% accuracy in identifying the most relevant gene panel. The underlying dataset includes standardised gene identifiers, disease annotations, MOI, and literature support, enabling seamless integration into bioinformatic pipelines.

Availability: The platform data is openly available at PanelAppRex base [Data set], Zenodo https://doi.org/10.5281/zenodo.15736689, with source code at https://github.com/DylanLawless/PanelAppRex, and demonstration page at https://panelapprex.github.io/landing_page. The dataset is maintained for a minimum of two years following publication.

Acronyms

AD Autosomal dominant . . . 9
AI Artificial intelligence . . . 6
AR Autosomal recessive . . . 9
API Application Programming Interface . . . 17
CSV comma-separated values . . . 7
GE Genomics England . . . 3
HGNC Human Genome Organisation Gene Nomenclature Committee . . . 4
IEI Inborn Errors of Immunity . . . 7
IEM Inborn Errors of Metabolism
HTML HyperText Markup Language . . . 5
JACI Journal of Allergy and Clinical Immunology . . . 6
MOI Mode of Inheritance . . . 4
OMIM Online Mendelian Inheritance in Man . . . 5
PDF Portable Document Format . . . 7
PID Primary Immunodeficiency . . . 7
RAG Retrieval-Augmented Generation . . . 4
RDS R Data Serialization format . . . 5
TSV Tab-Separated Values . . . 5

1 Introduction

Disease-gene panels are widely used in both clinical and research settings to support the diagnosis and interpretation of genetic disorders. These panels provide structured lists of genes known to be associated with specific phenotypes or disease groups, helping clinicians prioritise candidate variants during genomic analysis. Sources like Genomics England (GE)'s PanelApp and PanelApp Australia host comprehensive, expert-curated panels that are actively maintained to reflect new genetic knowledge (1). For instance, these resources are integral to the NHS National Genomic Test Directory and the 100,000 Genomes Project (1). Despite their utility, these panel datasets are distributed across multiple platforms and formats, and are difficult to aggregate programmatically without data loss or inconsistency. Manual selection, interpretation, and cross-referencing of gene panels remain labour intensive, especially when integrating with other variant-level annotations from genomic resources.

Accurate causal variant interpretation requires consistent integration of structured annotations from key resources such as GE's PanelApp, ClinVar, UniProt, and Ensembl (1–4). To address this, we developed PanelAppRex, a tool that automates panel aggregation and prepares a machine-readable dataset with complete coverage of core

gene-level fields, including gene identifiers, inheritance modes, disease terms, and supporting evidence. The dataset is suitable for integration into AI-based workflows and downstream variant interpretation pipelines. In parallel, PanelAppRex provides a natural language-style search interface that streamlines discovery by supporting queries based on gene names, phenotypes, disease groups, and other key attributes. This functionality is enhanced by evidence-based Retrieval-Augmented Generation (RAG), allowing users to explore curated gene panel information efficiently.

2 Methods

2.1 Data

The PanelAppRex core model contained 58,592 entries consisting of 52 sets of annotations, including the gene name, disease-gene panel ID, disease-related features, confidence measurements. Data from gnomAD v4 comprised 807,162 individuals, including 730,947 exomes and 76,215 genomes (5). This dataset provided 786,500,648 single nucleotide variants and 122,583,462 indels, with variant type counts of 9,643,254 synonymous, 16,412,219 missense, 726,924 nonsense, 1,186,588 frameshift and 542,514 canonical splice site variants. ClinVar data were obtained from the variant summary dataset available from the NCBI FTP site, and included 6,845,091 entries, which were processed into 91,319 gene classification groups and a total of 38,983 gene classifications (2). Data from Ensembl was sourced for validation of identifiers such as gene IDs and Human Genome Organisation Gene Nomenclature Committee (HGNC) symbols (4). Disease interactions were compared against GE's PanelApp (1).

2.2 Implementation

PanelAppRex was implemented in R and integrated data from the sources listed above. Reference data was merged for comparison with the core model in two formats: a simplified version (Panel ID, Gene) and a complex version including metadata such as confidence level, Mode of Inheritance (MOI), and disease information, and several metadata summary statistics. In addition, the tool incorporated a search module to execute complex user queries. The search functionality supports queries by gene names, phenotypes, disease names, disease groups, panel names, genomic locations and other identifiers. RAG was used to improve the natural queries in hidden states based on evidence about the disease and gene function by supplementing the data with additional sources including ClinVar, UniProt, etc.

2.3 Usage

We provide both a pre-compiled user interface and an analysis-ready dataset. On a desktop browser, the user interface allows for queries via the integrated search bar in

HyperText Markup Language (HTML), where a JavaScript function splits the query into individual terms and progressively filters all entries - retaining only those that match all active terms while ignoring unmatched ones. This enables users to perform complex, partial matching queries (e.g. "paediatric *RAG1* primary immunodeficiency skin disorder") to rapidly identify the panel most closely associated with their hypothesis on primary immunodeficiency and paediatric skin disorders. After querying and selecting a panel, the complete underlying data is returned.

The analysis-ready dataset contains the same dataset in multiple file formats. Our recommended strategy is a bioinformatic approach which can use the union of returned panels, or to apply a scoring system to rank panels in multi-group analyses. Bioinformatically, users can import the provided, ready-for-use, datasets in Tab-Separated Values (TSV) or R Data Serialization format (RDS) formats. A typical use case might involve users merging with their own omic data based on gene or Ensembl ID. The following code snippet, available in `minimal_example.R`, demonstrates how to load the data in R:

```
# TSV format
path_data <- "../data"
core_path <- paste0(path_data, "/PanelAppData_combined_core")
minimal_path <- paste0(path_data, "/PanelAppData_combined_minimal")

df_core <- read.table(
    file= paste0(core_path, ".tsv"),
    sep = "\t", header = TRUE)

df_minimal <- read.table(
    file= paste0(minimal_path, ".tsv"),
    sep = "\t", header = TRUE)

# Rds format
rds_path <- paste0(path_data, "/PanelAppData_combined_Rds")
df_core <- readRDS(file= rds_path)
```

2.4 Validation for completeness in core data

To ensure the reliability and completeness of the core dataset, we systematically assessed whether key gene-level fields were present for each entry after merging the core dataset. Specifically, we checked for non-missing values in gene symbols, associated publications, disease panel names, MOI, and Online Mendelian Inheritance in Man (OMIM) gene IDs. Where HGNC or Ensembl gene IDs were missing, we used programmatic queries to the Ensembl database via the '`biomaRt`' R package to recover the identifiers using the available HGNC symbol as input (6). This validation step was applied to the full integrated PanelAppRex dataset.

2.5 Benchmarking for manual queries

To mimic a clinician who is trying to diagnose a genetic disease, we systematically selected genetic diagnosis case reports that used exome or genome sequencing. We chose the first five eligible case reports from each of three sources: Journal of Allergy and Clinical Immunology (JACI), the journal Neurology, and for random cross-disciplinary topics we used pre-defined PubMed search restricted to the last 12 months and free full text, yielding a total of 15 case studies across phenotypes. This selection allowed us to validate the method across immunological, neurological, and other disease areas.

For each case report, we used only the clinical narrative before any genetic analysis was introduced in the original publication. We used a standardised method to select the phenotype terms from each case report and then rank by most to least importance or relevance. The ranked phenotype terms were then used a keyword queries, simulating a naïve starting position for a clinician. Search terms were added incrementally until no panels matched. We then tested whether our PanelAppRex tool could successfully retrieve panels that included the final causal gene reported in each case study. The sources, queries, and results are shown in **Table S1**.

2.6 Applications in AI-based RAG

To demonstrate use as part of Artificial intelligence (AI)-based RAG, we derived mechanistic text for each gene from UniProtKB human proteome and merged these summaries with PanelAppRex genes and panel metadata (panel name, disease group, mode of inheritance, and reported phenotypes) to create a single context dataset for each panel. We then applied a deterministic GPT-based model (gpt-4.1-mini, OpenAI; temperature 0.1) with a fixed prompt that requested a concise, clinician-facing description of shared mechanisms and phenotypes for the gene set. Across all processed panels, this involved approximately 7 million tokens and 2 million output tokens, corresponding to an input to output ratio of 0.261. The model produced three structured outputs per panel (a detailed analysis section, an overview paragraph, and 3 to 6 bullet points), which we stored together with panel identifiers as a high-level index that can be queried in downstream RAG workflows.

3 Results

3.1 Core dataset

PanelAppRex successfully aggregated data, currently from 451 panels and several genomics databases, to offer a database and user-friendly search functionality (**Figure**

1). Users can retrieve results filtered by gene names, phenotypes, disease groups and other criteria. The system returned a table view with panel details and provided options for exporting results in comma-separated values (CSV), Excel, or Portable Document Format (PDF). Bioinformatic uses may include generating virtual panels, constructing prior odds, or supporting formal reporting in qualifying variant protocols. Summary statistics are reported in **Figure S2**. The majority of genes were cited with high confidence. Panel reuse varied across genes, with some like *RAG1* appearing in up to 7 distinct panels (**Figure S2 B**). The majority of panels had fewer than 1000 genes. For example, panel ID 398 contains 572 genes associated with Primary Immunodeficiency (PID), also referred to as Inborn Errors of Immunity (IEI), while panel ID 1220 includes 1675 genes linked to IEI and unexplained death in infancy (**Figure S2 C**).

3.2 Validation of completeness in the core dataset

To assess whether the dataset was analysis-ready, we audited completeness of key annotation fields required for joining, filtering, and interpretation (**Figure S1**). For publications, disease panel names, MOI, and OMIM gene identifiers, we report entry-level completeness across all gene-panel entries. For HGNC and Ensembl gene identifiers, we used automated recovery of missing IDs from Ensembl as it is not a subjective decision. After recovery, the core identifiers required for gene-panel linking were present at 100% completeness. A small number of entries are incomplete for Ensembl ID such as *HPE1* because it is a phenotype identifier rather than gene name, and is thus not present on HGNC. Some secondary annotation fields remained incomplete such as MOI, and we did not attempt to fill them in the core dataset because this would require non-primary-source decisions which are better reserved for independent downstream implementations. The validated dataset therefore provides a stable and analysis-ready identifier layer for downstream integration (**Figure S1**).

3.3 Benchmarking for manual queries confirms accurate retrieval of causal disease genes

To evaluate the practical utility of our system on a desktop browser, we applied the benchmarking approach to fifteen published genetic diagnosis case studies that used exome or genome sequencing. These comprised five immunology-focused PID case reports, five neurology-focused case reports, and five randomly selected domains, as described in Methods. The query process retrieved gene panels using natural language-style input. For example, in the first case study on Hereditary Angioedema, the authors reported suspecting that the condition was linked to "*SERPING1* Factor XII and edema" which we used as the query terms. Although the true causal gene term *F12* (using its official HGNC name) was not mentioned, the inclusion of the alias name and/or related disease terms enabled the system to correctly retrieve the panel containing *F12* based on the hidden search knowledge.

While an expert might readily recognise "Factor XII" as an alias name for *F12*, this correct query result demonstrates that the system can make such connections automatically, based on its extensive metadata, supporting users with varying levels of genetic expertise. Most of the case report queries relied purely on phenotypic descriptions without any candidate genes.

Notably, for all case studies, all correct causal genes were successfully retrieved by the system despite never appearing explicitly in any of our submitted query terms (**Table S1**). In our evaluation, PanelAppRex returned panels in which the causal gene was present in 85.6% of all returned panels, with the subjective user-selected "best panel" achieving a perfect accuracy of 100% (**Figure 1**). These metrics were derived by comparing the causal gene identified from the case study abstracts to the panels returned by our query, and by applying a subjective relevance measure to mimic intuitive panel selection - acknowledging that certain panels, such as the established PID gene panel, are inherently more reliable than broader, less specific panels (**Table S1**).

Overall, the results confirm that our approach accurately identifies the most relevant panels and effectively supports clinical decision-making in complex diagnostic scenarios.

3.4 Applications in AI-based RAG

Across all panels, we used PanelAppRex as the basis for generating AI-derived summaries that captured panel-specific mechanisms and phenotypes, conditioned on the disease context and the unique combination of genes in each panel. These short structured overviews are exposed as a research-only info box in the interface, giving users rapid context before expanding the full panel and enriching the semantic query layer used for retrieval. In aggregate, the RAG step consumed approximately 7 million input tokens and produced 2 million output tokens while condensing the 6.6 million word gene knowledge base into about 135,000 words of panel-level summaries, corresponding to roughly fifty-fold compression of the raw text while retaining panel-specific mechanistic and phenotypic signal. The same protocol is easily modifiable to allow real-time RAG requests from users, either on from commercial provides like OpenAI API or local open-weight models inside protected research environments by simply redirecting the `system_prompt` and `writeLines` functions as shown in our code repository.

3.5 Structured inheritance annotations reveal overlapping gene-MOI relationships

To assess the distribution of inheritance annotations we summarised all unique gene-MOI combinations across 58,592 entries as displayed in **Figure S3**. Among 6,280

distinct genes, a total of 9,237 unique gene-MOI pairs were identified, reflecting instances where a gene appears with multiple inheritance modes across different panels. The most common annotations were Autosomal recessive (AR) and Autosomal dominant (AD), consistent with their prevalence in monogenic disease panels. These structured inheritance annotations underpin key applications enabled by PanelAppRex, including MOI-filtered search and model-based prior estimation.

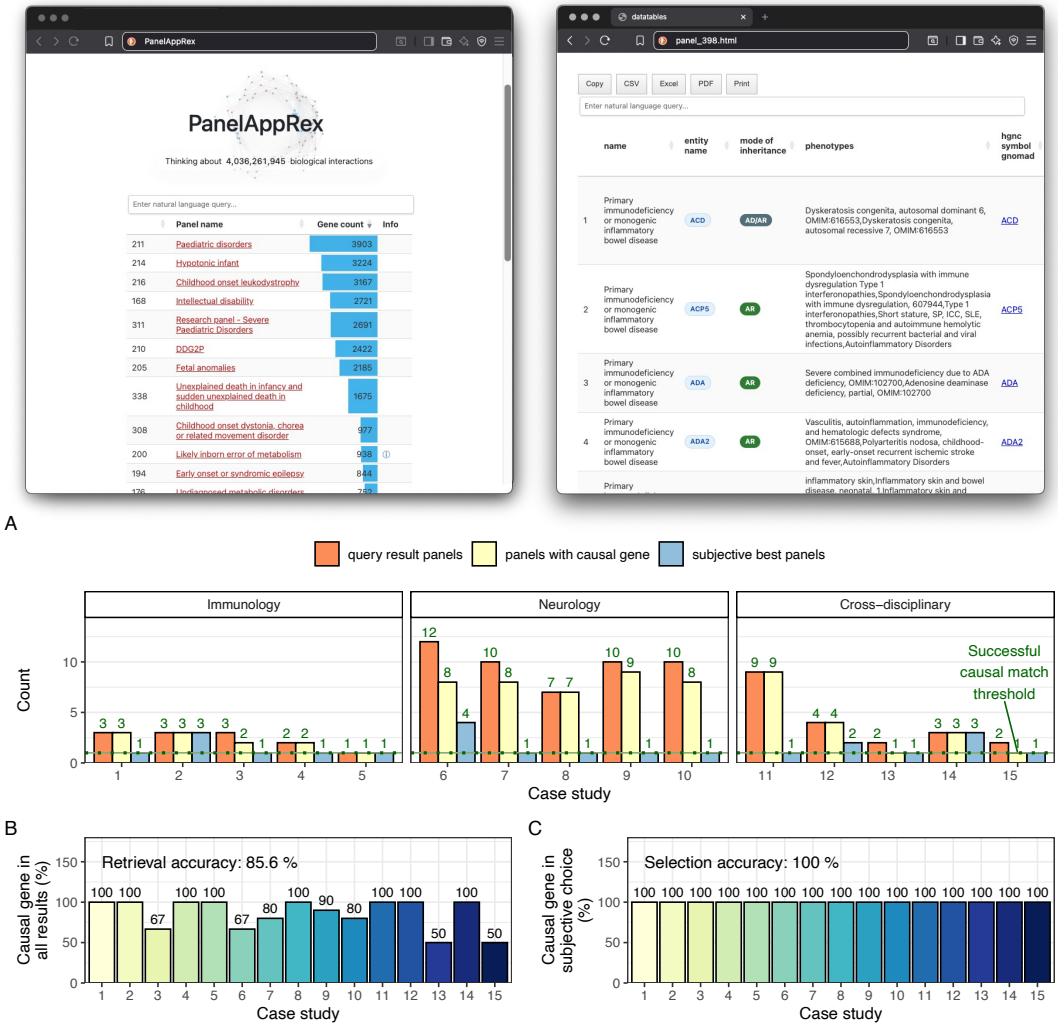

Figure 1: **PanelAppRex interface displaying search results for a complex query.** Top: screenshots showing the search interface, with the left panel displaying the full database before filtering and the right panel showing detailed results for a selected panel. Bottom: benchmark metrics are presented as follows: (A) for each of the fifteen case studies, the total number of panels returned after query input, the number of panels that included the causal gene, and the single best panel selected (always 1 by default); (B) the percentage of all returned panels that included the true causal gene; (C) the percentage of the single best panels that contained the true causal gene.

4 Discussion

PanelAppRex provides an accessible and validated platform for querying, aggregating, and exporting curated disease-gene panel data. Designed for both clinical and research use, it simplifies study planning and variant interpretation through a user-friendly interface, while also offering a harmonised dataset for programmatic workflows. The core model integrates over 58,000 panel-gene associations, with full coverage of key fields including HGNC symbols, Ensembl IDs, OMIM annotations, MOI, and supporting publications.

Beyond immediate usability, PanelAppRex lays a foundation for genome-wide statistical modelling of disease risk. With annotations for gene-disease relationships, the dataset provides a core building block towards rigorous estimation of the prior probability of observing variant classifications (e.g. benign, pathogenic) under different disease MOI. This addresses a key challenge in clinical genetics: the absence of principled priors for variant interpretation that incorporate not only known pathogenic variants (true positives), but also unobserved pathogenic variants (false negatives) and the absence of pathogenic findings (true negatives) (12; 13). By integrating high-resolution allele frequencies from gnomAD (5), curated classifications from ClinVar (2), and structured gene-disease associations (1), PanelAppRex can enable the derivation of calibrated genome-wide priors for Bayesian models of genetic risk.

These capabilities go beyond panel lookup to support evidence-aware diagnostics. Probabilistic models incorporating both observed and unobserved variation can quantify uncertainty, resolve ambiguous findings, and refine variant prioritisation, scaling naturally to support complex, multi-gene disorders. The dataset also provides a substrate for AI-driven approaches to variant interpretation, including probabilistic inference, reinforcement learning, and deep annotation pipelines (14; 15).

Several limitations should be acknowledged. Not all known coding genes are currently linked to a disease panel; we prioritised high-confidence, traceable annotations over broad but less reliable coverage. Some genes are over-represented across multiple panels due to historical research biases, leading to non-uniform panel enrichment. We performed benchmarks to measure success for both relatively simple gene-phenotype relationships (i.e., PID) and more complex relationships (i.e., neurogenetic disorders) but ultimately, a measure of long-term public option will be required to determine the practical benefit. Additionally, not all panels returned by queries may be equally informative. For example, during benchmarking, the COVID-19 research panel frequently appeared alongside the PID panel due to overlapping genes. While technically accurate, such panels may be less relevant to users focused on clinical diagnosis of immunodeficiency. In bioinformatic workflows, these choices can be refined systematically or excluded using filters or downstream logic. In case study three, for example, a disease-based query returned three panels, one of which did not contain the causal gene. This is expected, as broader classifications may include non-specific panels, hindering sensitivity. However, the query was still considered 100% successful, as the causal gene appeared in the combined result set, consistent with the recommended

bioinformatic strategy of using the union of returned panels. We note that a pre-compiled offline database is useful for secured computing environments but it misses out on the benefits of well maintained primary source APIs: for rapidly changing resources, direct API access can provide more frequent updates than an offline snapshot, which we cannot match in the same way.

PanelAppRex bridges expert-curated panel knowledge with genome-scale statistical reasoning. It offers a robust tool for interactive search, a validated dataset for programmatic access, and a scalable framework for quantitative genomic modelling. Future work will focus on expanding supported queries, integrating additional variant types and annotations, and enabling more sophisticated applications in variant interpretation and risk estimation.

Acknowledgements

ClinVar asks its users who distribute or copy data to provide attribution to them as a data source in publications and websites (2). The use of data from UniProt was based on Creative Commons Attribution 4.0 International (CC BY 4.0). We acknowledge GE for providing publicly accessible PanelApp data. The use of data from GE panelapp was for research purposes and is not for healthcare or commercial services. The use of GE panelapp source code was under the Apache License 2.0. The experimental (RAG) resource includes machine-assisted annotations generated using the OpenAI API according to their terms of use. We do not access OMIM data but instead direct readers using gene identifiers linked to OMIM.org record pages for further reading and acknowledge this resource. Ensembl data and code are available without restriction and software is provided under the Apache License 2.0 (4). Ensembl and its satellite sites conform to the EBI terms of use; as such we acknowledge attribution to EMBL-EBI for the use of its data resources and tools.

Contributions

DL designed and performed analyses and wrote the manuscript. SB, AS, SS, JF, LJS designed analysis and wrote the manuscript. The Quant Group is a collaboration across multiple institutions where authors contribute equally; the members on this project were DL, SB, and AS.

Competing interest

The authors declare no competing interest.

Ethics statement

This study only used data which was previously published and publicly available, as cited in the manuscript. This SwissPedHealth study, under which this work was carried out, was approved based on the advice of the ethical committee Northwest and Central Switzerland (EKNZ, AO_2022-00018). The study was conducted in accordance with the Declaration of Helsinki.

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Data availability

- Data: Lawless, D. (2025). PanelAppRex base Zenodo. https://doi.org/10.5281/zenodo.15736689
- Source code: https://github.com/DylanLawless/PanelAppRex
- Demonstration page: https://panelapprex.github.io/landing_page.

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5 Supplemental

Table S1: **Summary of case study queries and PanelAppRex results.** We recommend automated command-line use that accounts for all query results. For manual interface users, subjective decisions are naturally expected. We therefore measured "subjective best panels" as those reasonably preferable to the clinical query and unlikely to be excluded by users. In the benchmark scenario, broad or less relevant panels (e.g. "COVID-19 research") might be deprioritised in favour of more clinically aligned options such as "Fetal anomalies", or "Paediatric disorders". The results are summarised in **Figure 1**. Case study 5 had five individual cases and patient information was significantly longer than other studies. Therefore, we used OpenAI model o3-mini to converted each to a standardised keyword query automatically, thereby removing our subjective bias and aligning with the other queries.

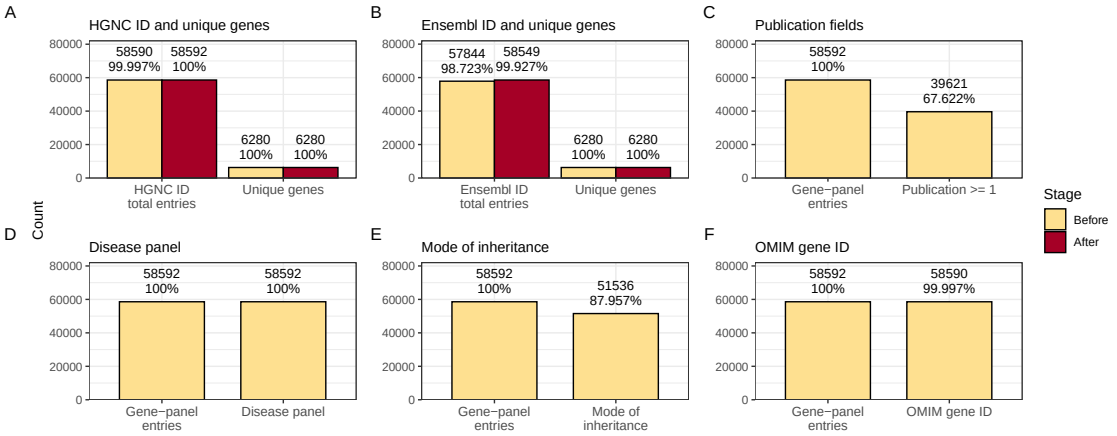

Figure S1: **Validation of completeness major annotation fields in the core dataset.** (A) Entry-level completeness of HGNC identifiers before and after recovery using Ensembl biomaRt with HGNC symbol input, shown together with the total number of unique genes in the dataset as a reference. (B) Entry-level completeness of Ensembl gene identifiers before and after recovery using Ensembl biomaRt with HGNC symbol input, shown together with the total number of unique genes in the dataset as a reference. (C) Entry-level completeness of publication annotations, shown as the number of gene-panel entries compared to those with at least one linked publication. (D) Entry-level completeness of disease panel annotations, shown as the number of gene-panel entries with a gene and the number with a disease panel name. (E) Entry-level completeness of mode of inheritance annotations, shown as the number of gene-panel entries with a gene and the number with a recorded mode of inheritance. (F) Entry-level completeness of OMIM gene identifier annotations, shown as the number of gene-panel entries with a gene and the number with an OMIM gene identifier. For all panels the percentages are calculated relative to total gene-panel entries, except for panels A and B where the additional "unique genes" bars are calculated relative to the total number of unique genes.

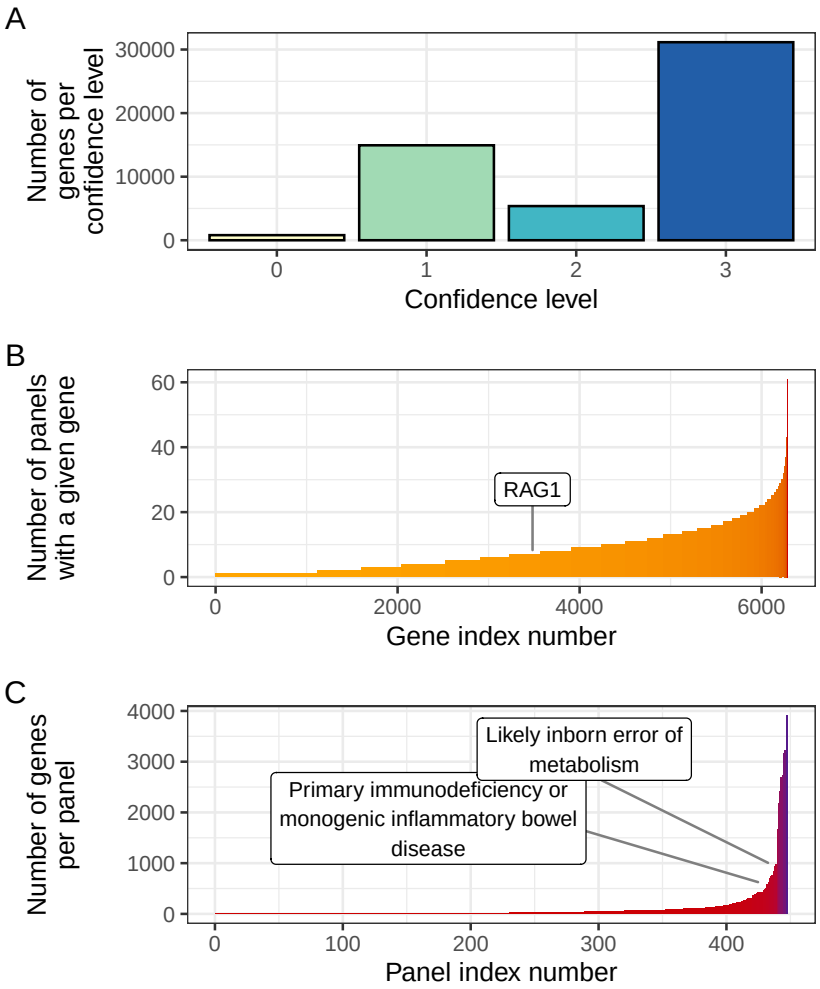

Figure S2: **Summary of gene confidence levels, reuse across panels, and panel sizes in the PanelAppRex dataset.** (A) Number of genes per confidence level. (B) Number of panels in which each gene is included, with the example gene *RAG1* highlighted to demonstrate that it is present in 7 panels. (C) Number of genes per panel, with two representative panels annotated: panel ID 398 (PID/IEI) and panel ID 1220 (IEM).

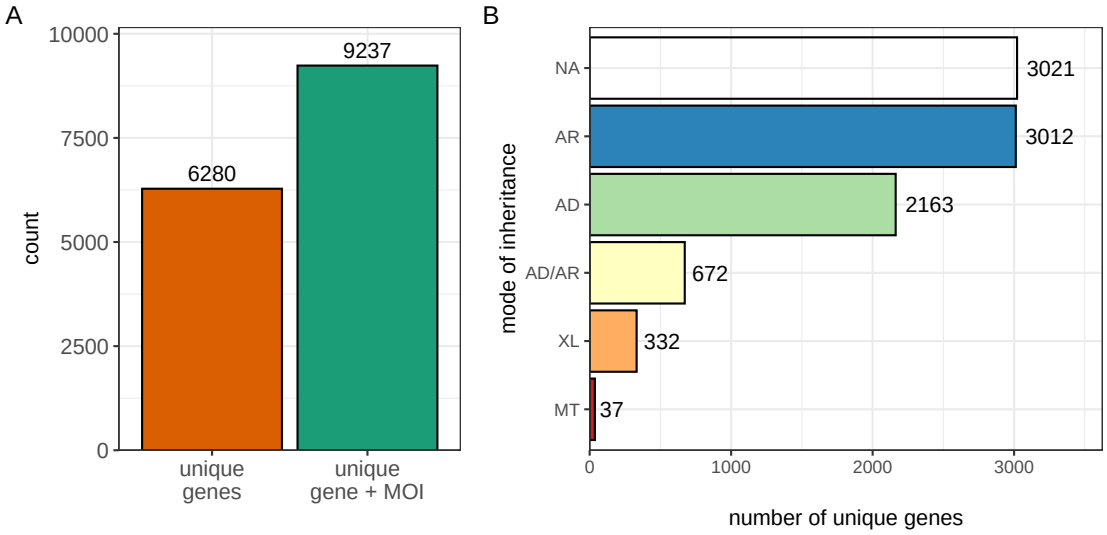

Figure S3: **Summary of mode of inheritance annotations in the PanelAppRex dataset.** (A) Counts of unique genes annotated with each MOI, based on non-redundant gene-MOI combinations. (B) Total number of unique genes and total number of gene-MOI combinations in the harmonised PanelAppRex dataset.

5.1 Performance

To assess whether PanelAppRex improves practical query performance beyond direct use of an upstream source, we performed a cold-start benchmark against a best-case GE PanelApp Application Programming Interface (API) workflow by re-using the first validation case query, "*SERPING1*, Factor XII, and edema" (**Figure S4**). We used the same query in both settings. This comparison is not fully like-for-like, because the GE PanelApp API does not accept the same phenotype-style query and return a ranked panel result directly. Instead, panel data must first be retrieved and assembled before ranking can begin. To make the comparison as favourable as possible to the API workflow, we deliberately included the known relevant panel from validation case study 1 and then increased the number of queried panels. In practice, a user would not usually know that panel in advance.

PanelAppRex local was faster at every tested subset size for time to final ranked result. The API workflow also required increasing numbers of panel-level requests in order to assemble the returned data before ranking could begin. After 150 queried panels, the GE API returned HTTP 429 rate-limiting errors, so the full panel set could not be evaluated beyond in this setting without adding a time delay. In addition, the API comparator returned only panel content, whereas PanelAppRex provided the pre-integrated local dataset together with the annotations used elsewhere in the platform such as gene function and disease details.

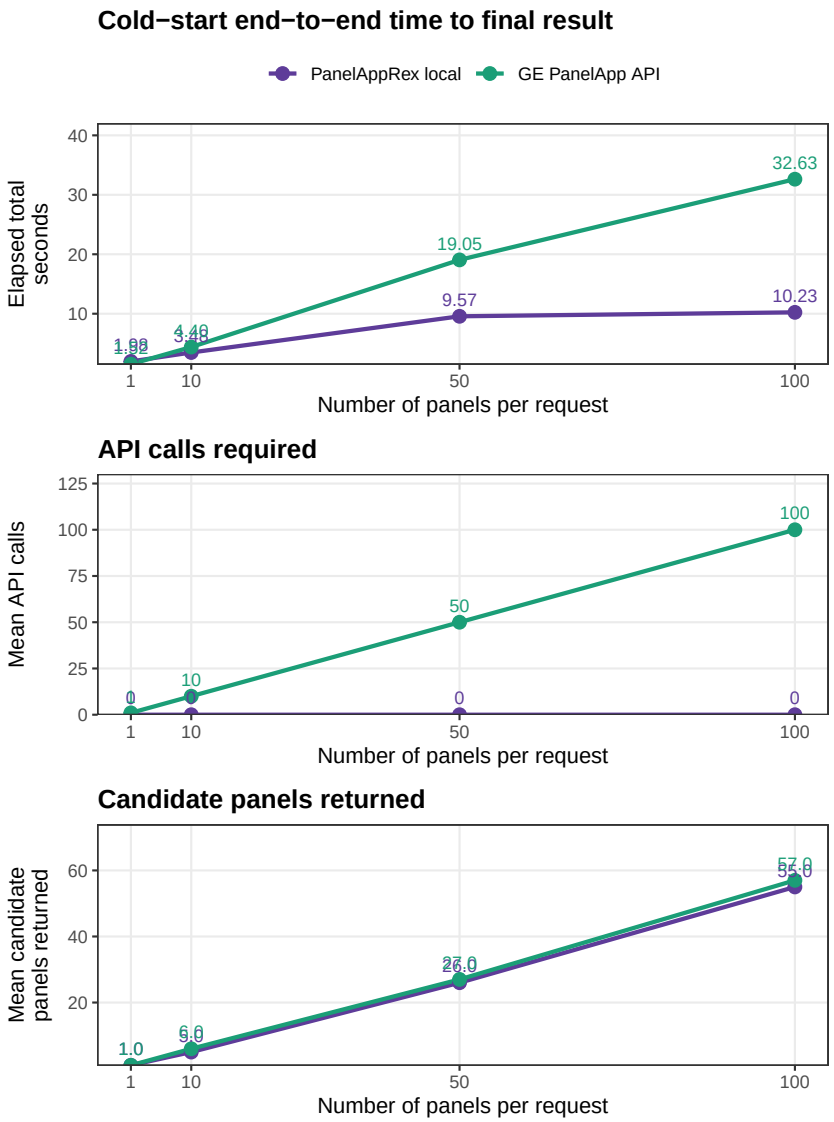

Figure S4: **Cold-start benchmark of PanelAppRex local against a best-case GE PanelApp API workflow.** The same clinical query from case study 1 was used in both settings. The x axis shows the number of panels in each benchmark set. For each run, the same growing panel set was used in both arms. The API comparator was deliberately made favourable by including the known relevant panel and then increasing the number of additional panels retrieved. (A) End-to-end elapsed time from query to final ranked result. (B) Mean number of API calls required to retrieve and assemble the queried GE panels before ranking. PanelAppRex local required no live API calls at query time. (C) Mean number of candidate panels returned after applying the input query to the retrieved panel set. For this speed benchmark, candidate panels were counted using at least one matching query term. After 150 panels in the benchmark set, the GE API returned HTTP 429 rate-limiting errors, so larger runs could not be completed in this benchmark.